MIL-H-965D (SHIPS)
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SUPERSEDING
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(see 6.2)

MILITARY SPECIFICATION

HEATERS, WATER, ELECTRIC, NAVAL SHIPBOARD

1. SCOPE

1.1 This specification covers water heaters which are stationary, self-contained, and electrically heated, for use aboard ships of the U. S. Navy.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116 - Preservation,Methods Of.
MIL-S-901 - Shock Tests, H. I.(High-Impact); Shipboard Machinery, Equipment and Systems, Requirements For.
MIL-F-1183 - Fittings, Tube, Cast Bronze, Silver-Brazing.
MIL-D-1000/2 - Drawings, Engineering and Associated Lists.
MIL-R-2033 - Relays for Naval Shipboard Electrical Service.
MIL-C-2174 - Controllers, Direct-Current, Naval Shipboard.
MIL-C-2212 - Controllers, Alternating-Current, Naval Shipboard.
MIL-V-13612 - Valves, Relief, Pressure and Temperature.
MIL-P-15024 - Plates, Identification ——— Information and Marking for Identification of Electrical, Electronic and Mechanical Equipment.
MIL-M-15071 - Manuals, Equipment and Systems.
MIL-P-15137 - Provisioning Technical Documentation for Repair Parts for Electrical and Mechanical Equipment (Naval Shipboard Use).
MIL-S-15291 - Switches, Rotary, Snap Action.
MIL-P-15328 - Primer, Pretreatment (Formula No. 117 for Metals).
MIL-W-16878 - Wire, Electrical, Insulated, High Temperature.
MIL-I-17244 - Indicators, Temperature, Direct-Reading Bimetallic, (3 and 5 Inch Dial).
MIL-F-20042 - Flanges, Pipe, Bronze (Silver Brazing).
MIL-H-22577 - Heating Elements, Electrical; Cartridge, Strip and Tubular Type.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-167 - Mechanical Vibrations of Shipboard Equipment.
MIL-STD-278 - Fabrication, Welding and Inspection of Machinery, Piping and Pressure Vessels for Ships of the United States Navy.
MIL-STD-769 - Thermal Insulation Requirements for Machinery and Piping.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)
ASME Boiler and Pressure Vessel Code, Section VIII - Unfired Pressure Vessels.

(Application for copies should be addressed to The American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, N.Y. 10017.)

NATIONAL BUREAU OF STANDARDS
Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20401.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Design. Water heaters shall consist of a pressure tank fitted with electric heating elements, a thermostatic switch, a pressure relief valve, thermometer, drain connection, drain valve (if required), fittings for connecting piping, a magnetic contactor, and "ON-OFF" switch. The water heater shall be so designed that the tank can be cleaned, and the heating elements and thermostatic switch readily removed without disturbing the insulation on the tank or removing the piping.

3.1.1 Sizes. Size of heaters shall be 5, 10, 25 and 50 gallon capacities, or in special capacities as specified (see 6.1).

3.1.2 Shock. Heaters shall be capable of withstanding the high-impact (H.I.) shock tests of 4.2.7 for grade B, class I, equipment as defined in MIL-S-901.

3.1.3 Vibration. Heaters shall be capable of withstanding the vibration test of 4.2.8 without mechanical damage or malfunctioning.

3.1.4 Welding and allied processes. Welding and allied processes shall be in accordance with MIL-STD-278.

3.1.5 Screw thread standards. Screw threads shall be in accordance with Handbook H28. The design shall utilize standard tools to the greatest extent possible. Standard tools are defined as those tools listed in the Federal Supply Catalog (copies of this catalog may be consulted in the office of the Government Inspector).

3.1.6 Tank. The tank shall be designed to withstand an internal hydrostatic test pressure of 150 pounds per square inch (p.s.i.) and continuous working pressure of 100 p.s.i. The shell shall be cylindrical and the material used shall be 90-10 copper nickel or copper silicon alloy. The thickness of the shell and of the heads shall be in accordance with the ASME Boiler and Pressure Vessel Code, Section VIII. Tanks shall be furnished with a base for deck mounting by means of four or more bolts. The minimum size of the bolt threads shall be 5/8-inch, 11 UNC. A support for the heating elements shall be provided opposite to the element mounting flange.

3.1.7 Heating elements. Heating elements shall be of the tubular type conforming to MIL-H-22577. The heating elements shall be mounted in a horizontal position near the bottom of the tank. The elements shall be secured to the element flange by brazing. The element flange shall be of the same material as the shell (see 3.1.6). The element gasket shall be of sheet asbestos.

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3.1.7.1 Wattage. The wattage rating shall be selected to give the energy required to heat the tank when filled with water at 40°F. to at least 130°F. within 1 hour, with rated voltage applied to the elements. Watt density shall not exceed 50 watts p.s.i. of element heating surface.

3.1.7.2 Voltage. The heaters shall be suitable for operation on either one of the following power supplies as specified (see 6.1):

(a) 440 volts, 3-phase, 60 cycles, alternating current (a.c.).
(b) 115 or 230 volts, direct current (d.c.).
(c) Special voltages.

3.1.7.3 Dielectric strength. The heaters shall satisfactorily withstand the dielectric strength tests specified in 4.2.4.

3.1.8 Thermostatic switches. Thermostatic switches shall be the direct connected immersion bulb type (without capillary tubing). Unless otherwise specified (see 6.1) thermostatic switches shall be adjustable to any temperature between the range of 90°F. to 165°F. The switch contacts shall close and open within plus or minus 10°F. of the thermostatic setting within the above temperature range. The temperature differential between the opening and closing of the contacts shall be not less than 5°F. nor more than 10°F. The switch shall be so located that at the time of opening of the contacts, the maximum temperature of the water at the top of the tank shall not exceed the actual switch operating temperature by more than 15°F. The thermostatic switches shall conform to MIL-C-2174 or MIL-C-2212, as applicable.

3.1.9 Controls. A magnetic contactor or relay and an "ON-OFF" switch operating in the control circuit shall be furnished with each water heater. The contactor or relay switch shall be mounted in a corrosion resisting steel enclosure that is insulated or located so that heat from the tank will not cause the operating temperature limits of the control devices to be exceeded. The enclosure shall have a removable, corrosion resisting steel cover for access to the control devices. Power leads shall not enter the enclosure through the cover plate. Nuts, screws and bolts shall be nickel-copper or corrosion resisting steel. The enclosure gasket shall be rubber.

3.1.9.1 For 440-volt, 3-phase, 60 cycle, a.c. heaters, a contactor conforming to MIL-C-2212 shall be used. Relays for d.c. heaters shall conform to MIL-R-2033. Momentary opening and closing of the contacts under shock is permissible where operation of the heater is not affected. The "ON-OFF" switch shall be a 10-ampere rotary switch conforming to MIL-S-15291.

3.1.10 Wiring. Wire used to connect the electrical components shall be type B, C, or D in accordance with MIL-W-16878. All wiring of the heater and controls shall be completed during manufacture so that only the connection of power leads is required at installation in a ship. The cable connection to the heater shall be as specified (see 6.1).

3.1.11 Relief valve. Each heater shall have a pressure relief valve installed in the top of the tank. The relief valve shall be type III in accordance with MIL-V-13612. The pressure relieving element shall relieve at 20 p.s.i. ± 30 p.s.i. above the shutoff pressure of the supply water pumps, as specified (see 6.1).

3.1.12 Thermometers. Each heater shall have a thermometer installed in the top of the tank. Thermometers shall be in accordance with MIL-I-17244, as follows:

(a) Dial size: - 3-inch.
(b) Case form: - Either back or bottom connected.
(c) Stem length: - As required.
(d) Range: Plus 20°F. to 240°F.

3.1.13 Piping and fittings. Inlet and outlet connections shall be either bronze brazing type unions in accordance with MIL-F-1183, or flanges in accordance with MIL-F-20042, as specified (see 6.1). The cold water inlet, if at the top of the tank, shall have a brass pipe extending to within approximately 2 inches of the bottom of the heater tank. A 3/4 inch iron pipe size flange shall be provided in the bottom of each tank for connection to a drain pipe. The drain piping and drain valve shall be furnished by the installing activity. Connections for inlet and outlet water piping, thermometer and relief valve shall extend above the heater lagging. Nipples shall be brass.

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3.1.14 Insulation. Heaters shall be thermal insulated and lagged in accordance with MIL-STD-769. The lagging shall not interfere with removal of the heater elements or thermostatic switch from the tank. The lagging shall be of corrosion resisting steel.

3.2 Painting. A primer pretreatment coating, formula 117 in accordance with MIL-P-15328, followed by one coat of Navy grey enamel, shall be applied to the external surfaces of the heater.

3.3 Onboard repair parts. Onboard repair parts shall be furnished as specified in table I for each size of heater installed.

Table I - Onboard repair parts.

Name of part	Quantity based on number of heaters installed.		
	1-5	6-10	11-20
Heater element assembly	1	1	2
Thermostatic switch assembly	1	2	4
Gaskets and seals	100 percent	100 percent	100 percent
"ON-OFF" switch	1	1	2
Contactor repair parts in accordance with MIL-C-2212 or MIL R-2033	As applicable		

3.3.1 Complete components and all serviceable parts including onboard repair parts covered in 3.3 shall be listed in accordance with MIL-P-15137.

3.4 Drawings. Drawings shall be types II and III in accordance with MIL-D-1000/2. Drawings shall be approved by the command or agency concerned. In addition to the data required by MIL-D-1000/2, drawings shall include the following:

(a) Type II drawings:

(1) Cross-sectional view of the heater assembly with enlarged details of construction.
(2) Dimensioned outline of heater assembly.
(3) Details of serviceable components.
(4) Wiring diagram.
(5) Operating electrical characteristics including number of heating elements, wattage of each, and total watts.
(6) Weight of heater assembly.
(7) Cable connection.

(b) Type III drawings:

(1) Capacity of heater in gallons.
(2) Temperature rise in 1 hour.
(3) Number of heating elements; wattage of each element and total watts.
(4) Electrical characteristics.
(5) Hydrostatic test pressure and working pressure.
(6) Reference to "ON-OFF" switch, thermostatic switch and contactor drawings.

3.4.1 Drawings of "ON-OFF" switch, thermostatic switch and magnetic contactors shall be furnished in accordance with the applicable component specification.

3.5 Manuals. Manuals shall be type III in accordance with MIL-M-15071 in the quantities specified (see 6.1).

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3.6 Identification and information plates. Each heater shall have an identification plate in accordance with types A, B, or C of MIL-P-15024, mounted on the terminal box or on the heater lagging. The plate shall contain the following data:

(a) Electric water heater.
(b) Nominal capacity in gallons.
(c) Voltage, wattage, phases and amperes.
(d) Federal stock number (if available).
(e) Serial number and drawing number of heater.
(f) Name and address of manufacturer.

3.7 Workmanship. Heaters shall be free from defects affecting their appearance or their operation. Mating surfaces of flanges and mounting pads shall be machined true.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First unit tests. One heater of each size offered for delivery on the first contract or order shall be subjected to the first unit tests specified in 4.2.2 through 4.2.8.

4.2.1 If the command or agency concerned is satisfied that the heaters conform to this specification, a waiver may be granted to omit first unit tests on subsequent contracts or orders (see 6.1). First unit tests shall be made after any change in design which affects performance characteristics. If production test data reveal variations beyond the normal manufacturing tolerance, complete first unit tests shall be made on that particular size of heater to demonstrate that it conforms to this specification.

4.2.2 Hydrostatic test. The complete heater shall be tested at a hydrostatic pressure of 150 percent of the design operating pressure, but in no case at less than 100 p.s.i.

4.2.3 Calibration. The calibration of the thermostatic switch shall be checked by a controlled temperature liquid bath system, or by thermocouples attached to the sensing bulb of the temperature control to indicate the temperature at which the switch contacts open at various settings within the specified range.

4.2.4 Dielectric tests.

4.2.4.1 General. The dielectric tests shall be made after all other tests have been completed. The test voltage shall be equal to twice the rated value of the terminal voltage plus 1000 volts. The frequency shall be not less than 60 cycles, and the wave shape shall approximate a sine wave. The test voltage shall be applied continuously for a period of not less than 1 minute.

4.2.4.2 Measurements of test voltage. The measurements of the voltage used in dielectric tests shall be made by means of a voltmeter whereby the meter derives its voltage directly from the high voltage circuit, either by means of a voltmeter coil placed in the testing transformer or through an auxiliary potential transformer.

4.2.4.3 Points of application. The test voltage shall be successively applied between each electric circuit and the frame with all other electric circuits and metal parts grounded.

4.2.5 Performance test. The heater shall be filled with water at approximately 40°F., temperature recorded and the thermostatic switch set to cut off at approximately 130°F. Rated voltage shall be applied to the heating circuit, and the water heated until the thermostatic switch automatically cuts off. The water temperature at the top of the tank, and the time elapsing until the switch cuts off, shall be recorded.

4.2.6 Temperature test. The control equipment shall be subjected to a 4-hour operation at the voltage specified in the contract or order to determine that the temperature rise of the current carrying parts does not exceed the maximum values permitted in MIL-C-2212 and MIL-C-2174.

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4.2.7 Shock test. The heater shall be subjected to the type A, H.I. shock test in accordance with MIL-S-901.

4.2.7.1 Disposal of shock tested heaters. Disposal of shock tested heaters shall be as follows:

(a) Heaters which have been subjected to H.I. shock tests and have failed to meet the requirements of this specification shall not be acceptable, either in whole or any of their parts on any contract or order.
(b) Heaters which have been subjected to the H.I. shock test and have successfully passed the test shall be reinspected in accordance with 4.2.2, 4.2.3, 4.2.4 and 4.4.1.

4.2.8 Vibration test. The heater shall be subjected to type I vibration test in accordance with MIL-STD-167.

4.3 Sampling for production inspection.

4.3.1 Lot. A lot shall consist of all heaters of the same size offered for delivery at one time.

4.3.2 Sampling for examination. A random sample of heaters shall be selected from each lot in accordance with MIL-STD-105 at inspection level II for the examination of 4.4. The acceptable quality level (AQL) shall be not more than 1.5 percent defective.

4.3.3 Sampling for production tests. A random sample of heaters shall be selected from each lot in accordance with MIL-STD-105 at inspection level II for the tests specified in 4.5. The AQL shall be not more than 1.5 percent defective.

4.4 Examination.

4.4.1 Heaters. Each heater selected in accordance with 4.3.2 shall be thoroughly examined to ascertain that the workmanship and design are in conformance with this specification.

4.4.2 Repair parts. All repair parts shall be carefully examined to determine that the workmanship and sizes are identical to those used in the heaters.

4.5 Production tests. Each heater selected in accordance with 4.3.3 shall be tested in accordance with 4.2.2, 4.2.3 and 4.2.4.

5. PREPARATION FOR DELIVERY

(Preparation for delivery requirements specified herein apply only to direct Government procurement. Preparation for delivery requirements between contractors and sub-contractors shall be as specified in the individual order.)

5.1 Domestic shipment and early equipment installation and for storage of onboard repair parts.

5.1.1 Heater.

5.1.1.1 Preservation and packaging. Preservation and packaging which may be the supplier's commercial practice, shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation.

5.1.1.2 Packing. Packing shall be accomplished in a manner which will insure acceptance by common carrier at the lowest rate and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules, or other carrier regulations, as applicable to the mode of transportation and may conform to the suppliers commercial practice.

5.1.1.3 Marking. Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, contractor's name and destination.

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5.1.2 Onboard repair parts.

5.1.2.1 Packaging. Nonferrous parts such as thermostatic elements, switch assemblies and so forth shall be unit protected and packaged in accordance with method III of MIL-P-116. All openings in piping units, and the like, shall be plugged or sealed to prevent entrance of foreign materials. Gaskets shall be packaged flat and provided with fiberboard stiffeners to prevent deformation. Parts shall be individually packaged except when used in sets.

5.1.2.2 Packing. Packing shall be accomplished in a manner which will insure acceptance by common carrier at the lowest rate and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules, or other carrier regulations as applicable to the mode of transportation and may conform to the suppliers commercial practice.

5.1.2.3 Marking. In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

5.2 Domestic shipment and storage or overseas shipment. The requirements and levels of preservation, packaging, packing and marking for shipment shall be specified by the procuring activity (see 6.1).

(5.2.1 The following provides various levels for protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made.

5.2.1.1 Preservation and packaging.

5.2.1.1.1 Level A. The heaters shall be preserved and individually packaged by method III of MIL-P-116. All openings to the heater shall be sealed to prevent entrance of foreign material.

5.2.1.2 Packing.

5.2.1.2.1 Level A. Heaters, packaged as specified, shall be individually packed in boxes conforming to any one of the following specifications at the option of the supplier:

Specification	Type or class
PPP-B-601	Overseas type
PPP-B-621	Class 2

5.2.1.2.2 Level B. Heaters, packaged as specified, shall be individually packed in boxes conforming to any one of the following specifications at the option of the supplier:

Specification	Type or class
PPP-B-601	Domestic type
PPP-B-621	Class 1

5.2.1.2.3 Boxes shall be modified by the addition of skids in accordance with the applicable box specifications. Cushioning, anchoring, blocking and bracing shall be in accordance with MIL-STD-1186. Box closure shall be as specified in the applicable box specification or appendix thereto.

5.2.1.3 Marking. In addition to any special marking required in the contract or order, marking of packages and shipping containers shall be in accordance with MIL-STD-129.)

6 NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size required (see 3.1.1).
(c) Voltage required (see 3.1.7.2).
(d) Temperature range of thermostatic switches if other than as specified in 3.1.8.
(e) Cable connections required (see 3.1.10.)

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(f) Relief valve setting (see 3.1.11).
(g) Whether unions or flanges are required (see 3.1.13).
(h) Number of manuals required (see 3.5).
(i) Whether first unit tests are to be conducted (see 4.2.1).
(j) Preservation, packaging, packing and marking requirements if other than as specified in 5.1 (see 5.2).

6.2 CHANGES FROM PREVIOUS ISSUE. THE EXTENT OF CHANGES (DELETIONS, ADDITIONS, ETC.) PRECLUDE THE ANNOTATION OF THE INDIVIDUAL CHANGES FROM THE PREVIOUS ISSUE OF THIS DOCUMENT.

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